Validation of the Automated Citation Classification Framework

Overview

To evaluate the robustness and reliability of the automated citation-mining framework, two independent validation procedures were conducted:

1. Binary document-level validation, assessing whether publications were correctly classified as microbiology-related or non-microbiology-related.
2. Multi-label keyword-level validation, assessing the agreement between automated and manual assignments across four predefined classification dimensions.

All validation procedures were implemented in a Jupyter Notebook (**explore_publications_manual_validation.ipynb**) using fixed random seeds to ensure deterministic sampling and fully reproducible results.

Binary Document-Level Validation

To evaluate document-level classification (microbiology-related vs. non-microbiology-related)

Sampling Strategy

A validation dataset was constructed as follows:

- 5% of entries from **Supplementary Data 4** (microbiology-related citations)
- 5% of entries from **Supplementary Data 3** (all citations)

Title normalisation was used to ensure exact matching between tables. This resulted in a combined validation set of 549 publications.

Sampling and dataset construction were fully reproducible via fixed random seeds.

Evaluation Procedure

To evaluate whether the automated system correctly distinguishes microbiology-related from non-microbiology-related citations, binary classification metrics were computed.

For each manually reviewed publication:

- Positive class: microbiology-related
- Negative class: non-microbiology-related

The following were computed:

- True Positive (TP): correctly identified microbiology-related publications
- False Positive (FP): non-microbial publications incorrectly labeled as microbial
- False Negative (FN): microbial publications missed by the classifier
- True Negative (TN): correctly identified non-microbial publications

Precision, recall, F1-score, and overall accuracy were calculated.

Results

From 549 manually evaluated publications, documented in **Supplementary Data 11**:

Metric	Value
TP	134
FP	19
FN	38
TN	358
Precision	87.6%
Recall	77.9%
F1-score	82.5%
Accuracy	89.6%

The high precision indicates that when the classifier assigns microbiology-related status, it is correct in the vast majority of cases. The lower but substantial recall indicates a conservative tendency, with some microbiology-relevant publications not being detected.

False positives were limited, suggesting minimal contamination of the microbiology-related corpus, while false negatives indicate that borderline or implicitly microbiology-relevant studies may occasionally be excluded.

Multi-label Keyword-Level Validation

To evaluate keyword-level classification across four predefined classification dimensions

Sampling Strategy

A random subset of 1% of **Supplementary Data 4** (microbiology-related citations until the end of 2025) was selected using a fixed seed (seed = 42). This resulted in 30 publications for manual review, which are listed in **Supplementary Data 12**.

Classification dimensions

Each publication was classified across four dimensions:

1. Targeted Organisms
2. Technical Targets
3. Methods
4. Topics

Each dimension consists of a predefined keyword universe.

Because publications may contain multiple keywords within a dimension, evaluation was conducted at the keyword level, not at the binary category level.

Evaluation Procedure

For each publication and each dimension:

- Let A be the set of automated keywords.
- Let M be the set of manually assigned keywords.
- Let U be the predefined keyword universe for that dimension. These keywords are predefined for every dimension/category, shown in details in **Supplementary Figure 5**.

The following were computed:

- True Positive (TP) = $|A \cap M|$, keywords present in both automated and manual annotations
- False Positive (FP) = $|A - M|$, keywords assigned automatically but not confirmed manually
- False Negative (FN) = $|M - A|$, keywords present in manual annotation but missed by the automated classifier
- True Negative (TN) = $|U - (A \cup M)|$, keywords assigned by neither approach

This set-based approach allows partial agreement. For example, if the automated classifier assigns multiple keywords and only some are manually confirmed, both TP and FP are recorded.

Category-level performance was obtained by summing TP, FP, FN, and TN across all publications.

Precision, recall, and F1-scores were computed per category.

Micro- and Macro-Averaging

Two averaging strategies were applied:

- Macro-average: arithmetic mean of per-category metrics (equal weight per dimension).
- Micro-average: metrics computed by pooling TP, FP, and FN across all dimensions (weighted by the total number of keyword decisions).

Macro-averaging reflects balance across classification dimensions, whereas micro-averaging reflects overall system behavior.

Results

The automated classifier achieved:

- Micro-averaged F1-score: 63.4%
- Macro-averaged F1-score: 68.9%

Category-specific performance:

Category	Precision	Recall	F1-score
Topics	87.5%	89.7%	88.6%
Methods	93.1%	77.1%	84.4%
Targeted organisms	75.6%	63.0%	68.7%

Technical target	61.1%	23.4%	33.8%

Performance was strongest for Topics and Methods, moderate for Targeted organisms, and lowest for Technical targets, primarily due to reduced recall in that category.

Reproducibility

All validation procedures were implemented in a Jupyter Notebook available on GitHub.

This notebook:

- Performs reproducible random sampling using fixed seeds,
- Constructs validation subsets,
- Normalises keyword labels,
- Computes TP, FP, FN, TN at the keyword and document level,
- Generates summary statistics and exportable result tables.

Manually validated sampling tables are saved as TSV files to ensure transparency and auditability (https://github.com/usegalaxy-eu/microbiology_galaxy_lab_paper_2026/tree/main/docs/supplementary).

- Binary document-level manual validation of microbiology-related citation classification: **Supplementary Data 11**
- Multi-label keyword-level manual validation of automated microbiology citation classification: **Supplementary Data 12**

Conclusion

The combined validation results demonstrate that:

1. The automated framework reliably distinguishes microbiology-related literature from unrelated citations with high precision and strong overall F1 performance.
2. High-level thematic and methodological classification (Topics and Methods) is robust and consistent.
3. Fine-grained technical target assignments remain more challenging, primarily due to reduced recall.

These results support the use of the automated framework for large-scale trend analysis and ecosystem mapping, while identifying specific areas where further refinement may improve label sensitivity.